

Functional Requirements Document

Digital STEM Content Delivery Platform

Project Details

Field	Details
Project Name	Digital STEM Content Delivery Platform
Client	ABC School
Document Type	Functional Requirements Document
Prepared By	Business Analyst
Version	1.0
Date	xx June 20xx

1. Document Purpose

This Functional Requirements Document defines the functional behaviour required for the Digital STEM Content Delivery Platform. It converts approved business requirements into system-level requirements that can be used by the Developer Team, UI/UX Designer, QA Tester and Product Owner during delivery.

2. Document Inputs

- Project Intake Note
- Stakeholder Register
- Discovery Questions and Discovery Meeting Notes
- As-Is Process Flow
- To-Be Process Flow
- Approved Business Requirements Document

3. Scope Summary

The platform will support teacher-led classroom usage. Teachers will log in through a web portal, access approved STEM and Coding content, filter content by class, subject and topic, and launch selected content in classroom display mode. Admin Staff will manage content records and content statuses. Principal, authorised Subject Teacher or Subject Matter Expert will review and approve learning content before it becomes visible to Teachers.

4. User Roles

Role	Functional Responsibility
Admin Staff	Creates and manages content records, metadata and content status.
Teacher	Views approved content, filters the library and launches classroom display mode.
Principal / Approval Owner	Reviews scope priority and approves content or approval rules.
Subject Teacher / SME	Reviews academic accuracy of learning content before approval.

5. Functional Requirements

5.1 Access, Library and Content Management Requirements

Req ID	Feature Area	Functional Requirement	Priority
FR-01	User Login	System shall allow Admin Staff, Teacher and Principal / Approval Owner to log in using valid credentials.	Must Have
FR-02	Login Validation	System shall validate user credentials before allowing access to the platform.	Must Have
FR-03	Role-Based Landing Page	System shall display relevant landing page options based on the logged-in user role.	Must Have
FR-04	Teacher Content Library	System shall allow Teachers to view the	Must Have

		STEM and Coding content library after login.	
FR-05	Approved Content Visibility	System shall display only Approved content to Teachers.	Must Have
FR-06	Content Filter	System shall allow Teachers to filter content by class, subject and topic.	Must Have
FR-07	Content Type Display	System shall display content type such as 3D video, animated video, coding simulation, visual explanation or interactive web content.	Should Have
FR-08	Content Details	System shall allow Teachers to view selected content details before launching it.	Must Have
FR-09	Classroom Display Mode	System shall allow Teachers to launch selected content in classroom display mode.	Must Have
FR-10	Display Mode View	System shall hide unnecessary admin controls in classroom display mode.	Must Have
FR-11	Admin Content List	System shall allow Admin Staff to view a list of content records.	Must Have
FR-12	Add Content Record	System shall allow Admin Staff to add a new content record.	Must Have

5. Functional Requirements

5.2 Status, Approval and Display Requirements

Req ID	Feature Area	Functional Requirement	Priority
FR-13	Edit Content Record	System shall allow Admin Staff to edit existing content records.	Must Have
FR-14	Mandatory Metadata	System shall require class, subject, topic, content type and status for each content record.	Must Have
FR-15	Learning Objective	System shall allow Admin Staff to add a learning objective for each content record.	Should Have
FR-16	Content Status	System shall support Draft, Under Review, Approved and Inactive content statuses.	Must Have
FR-17	Submit for Review	System shall allow Admin Staff to submit Draft content for review.	Must Have
FR-18	Under Review Visibility	System shall keep Under Review content hidden from Teachers.	Must Have
FR-19	Content Review	System shall allow Principal, authorised Subject Teacher or Subject Matter Expert to review content submitted for approval.	Must Have
FR-20	Approve Content	System shall allow authorised approver to mark content as Approved.	Must Have
FR-21	Approved Content Publishing	System shall make Approved content visible	Must Have

		in the Teacher content library.	
FR-22	Inactive Content	System shall hide Inactive content from Teacher view.	Must Have
FR-23	Correction Flow	System shall allow content that is not approved to be sent back for correction.	Should Have
FR-24	Error Message	System shall show a clear message if content cannot be opened.	Should Have

6. Functional Flow by Role

6.1 Teacher Flow

1. Teacher opens the web portal.
2. Teacher logs in using valid credentials.
3. System validates Teacher login.
4. Teacher opens STEM and Coding content library.
5. System displays only Approved content.
6. Teacher filters content by class, subject and topic.
7. System displays matching content.
8. Teacher selects content item.
9. Teacher clicks Launch Content.
10. System opens classroom display mode.
11. Teacher displays content on classroom screen, projector or smart board.

6.2 Admin Flow

12. Admin Staff logs in to the web portal.
13. Admin Staff opens content management screen.
14. Admin Staff adds or edits content record.
15. Admin Staff enters content metadata such as class, subject, topic and content type.
16. Admin Staff saves content in Draft status.
17. Admin Staff submits content for review.
18. System changes content status to Under Review.

6.3 Approval Flow

19. Principal, authorised Subject Teacher or Subject Matter Expert opens content for review.
20. Reviewer checks academic accuracy, learning objective, labels and content suitability.
21. If content is approved, System changes status to Approved.
22. Approved content becomes visible to Teachers.
23. If content is not approved, System sends content back for correction.

7. Data Fields Required

Field	Description	Owner	Mandatory
Content ID	Unique system generated content identifier	System	Yes
Content Title	Name of the learning content	Admin Staff	Yes
Class	Class or grade level	Admin Staff	Yes
Subject	Subject such as Physics, Chemistry, Mathematics or Computer Science	Admin Staff	Yes
Topic	Specific topic such as H2O molecule, loops or 3D geometry	Admin Staff	Yes
Content Type	3D video, animated video, coding simulation, visual explanation or interactive web content	Admin Staff	Yes
Learning Objective	What the content should help students understand	Admin Staff / Subject Teacher	Should be required
Content Status	Draft, Under Review, Approved or Inactive	System / Admin / Approver	Yes
File or Link	Content file path or content URL	Admin Staff / Content Team	Yes
Approval Owner	Person responsible for approval	Product Owner / Admin	Yes

8. Validation Rules

- User must enter valid login credentials to access the platform.
- Class, subject, topic, content type and status must be captured before content can be submitted for review.
- Only authorised users can approve content.
- Draft, Under Review and Inactive content must not be visible to Teachers.
- Only Approved content can appear in the Teacher content library.
- Classroom display mode must be available only for approved content.

9. Functional Assumptions

- User roles and login accounts will be available before testing.
- Sample content records will be available for development and QA testing.
- Final content file storage approach will be confirmed with the Developer Team.
- Principal / Product Owner will confirm authorised approval users.

10. Out of Scope for FRD

- Student login and student portal.
- Student progress tracking.
- Parent login.
- Payment gateway.
- Attendance features.
- AR-based and VR headset-based learning.
- Complex AI recommendation engine.

11. Next Step

After FRD review, the Business Analyst will prepare the Non-Functional Requirements Document, Business Rules Document, Use Case Document, user stories and acceptance criteria for Sprint 1 and Sprint 2 backlog preparation.